Online Food Delivery Analysis Dashboard

An end-to-end analytical pipeline processing 100,000 delivery records using Python, MySQL, and Streamlit. This project demonstrates technical proficiency in data engineering whilst extracting actionable business insights from real-world operational data.

Made with GAMMA

Project Overview & Technical Architecture

Project Objective

Building an analytical pipeline to process and visualize 100,000 delivery records, transforming raw data into strategic business intelligence

Technology Stack

MySQL 8.0.44 for data storage, Python with Pandas and SQLAlchemy for ETL, Streamlit for interactive visualization

Key Deliverable

Dynamic dashboard enabling real-time querying of financial performance, delivery efficiency, and customer satisfaction metrics

Data Architecture

Raw data contained missing values in critical columns including Peak_Hour and Cancellation_Reason. Python implemented categorical imputation strategies whilst converting data types for optimized SQL processing.

Import Efficiency

Chunksize parameters enabled large dataset import to DESKTOP-10RKQQ7 server without timeouts, demonstrating production-ready data engineering practices.

Made with GAMMA

Financial & Revenue Analysis

01

Monthly Revenue Trends

Formatted raw string dates into structured YYYY-MM formats, identifying seasonal peaks and troughs in revenue generation

02

Promotion Impact Analysis

Examined discount strategies against average profit margins, determining whether promotional campaigns deliver genuine value

03

High-Value Segments

Generated "High-Revenue Cities & Cuisines" report using FORMAT() for professional currency display, highlighting top-performing combinations

Made with GAMMA

Delivery & Restaurant Performance

Logistics Efficiency

Correlated Distance_km with Avg_Time across cities, identifying delivery zones with superior operational performance and route optimization opportunities

Quality Control Metrics

Built "Cancellation Rate by Restaurant" tracking system, flagging vendors impacting user experience and requiring operational intervention

Customer Satisfaction

Established direct correlation between Delivery_Rating and delivery speed, proving faster deliveries significantly increase customer ratings

Made with GAMMA

Operational Intelligence Insights

Peak Demand Visualization

Cleaned Peak_Hour data revealed order spikes throughout the day, suggesting optimal staffing windows to maximize delivery capacity during high-demand periods.

Revenue Protection Analysis

Quantified "Total Revenue Lost" by specific cancellation reasons, distinguishing between late delivery penalties versus wrong item replacements, enabling targeted operational improvements.

Made with GAMMA

Business Value & Scalability

Modular System Design

The architecture remains modular and extensible; new queries added to the query_bank dictionary trigger instant dashboard updates without code refactoring

Production-Ready Performance

SQL-level data formatting ensures the dashboard remains fast and responsive even with 100,000+ rows, filtering at the database layer rather than loading into Python memory

Strategic Decision Support

Transformed raw operational data into actionable intelligence, enabling data-driven decisions on staffing, vendor management, and promotional strategy

Made with GAMMA

Technical Implementation Details

Data Import

Chunksize parameters prevented server timeouts during large dataset loading to DESKTOP-10RKQQ7 infrastructure

Data Cleaning

Python categorical imputation filled missing values in Peak_Hour and Cancellation_Reason columns whilst maintaining data integrity

Type Conversion

Transformed Peak_Hour into binary 1/0 flags, enabling faster SQL processing and efficient boolean logic operations

Made with GAMMA

Key Technical Achievements

100K

Delivery Records

Processed 100,000 rows of operational data

15+

SQL Queries

Developed comprehensive query bank

3

Technology Stack

MySQL, Python, Streamlit integration

Notable Implementation Details

- Escaped percent signs (%%) in Python to enable SQL date format processing
- Custom UI header with responsive formatting for professional presentation
- Database-level filtering prevents 100,000 rows loading into Python memory
- FORMAT() function ensures currency displays professionally across all reports

Made with GAMMA

Strategic Impact Summary

Operational Efficiency

Delivery route optimization and staffing recommendations based on peak demand patterns

Vendor Management

Cancellation rate tracking enables proactive intervention with underperforming restaurants

Revenue Protection

Quantified revenue loss by cancellation reason, prioritizing operational improvements with highest ROI

Made with GAMMA

Key Takeaways

1 End-to-End Pipeline

Demonstrated full-stack data engineering from raw data import through SQL transformation to Streamlit visualization

2 Business Value Focus

Translated technical metrics into strategic insights: revenue protection, operational efficiency, customer satisfaction

3 Scalable Architecture

Modular design enables rapid iteration; new queries deploy instantly without system refactoring

4 Production-Ready Practices

Chunked data loading, SQL-level filtering, and efficient type conversions ensure performance at scale

Made with GAMMA